The Geometric Affective Survey: Operationalizing Three-Axis Emotional Measurement

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Current survey instruments measure emotional experience on a single axis. The Likert scale maps responses from 1 to 5 or 1 to 10. The PHQ-9 measures frequency of symptoms. The Net Promoter Score produces a single number between –100 and 100. All of these instruments collapse multidimensional emotional experience into one dimension. This paper proposes the Geometric Affective Survey (GAS), a three-axis instrument that maps responses to valence (positive-negative), activation (calm-excited), and intensity (mild-strong) relative to a respondent-specific baseline (sigma). The instrument produces a coordinate in three-dimensional affective space rather than a number on a line. Responses can be visualized as point clouds, clustered, and tracked over time. This paper describes the instrument design, its theoretical basis in a three-dimensional emotional geometry formalized across four companion papers (Riggleman 2026j, 2026k, 2026l, 2026m), and demonstrates backward compatibility on two independent datasets: the IBM HR Analytics dataset (n=1,470), where the geometric decomposition reveals a 2.2x difference in attrition rates across geometric quadrants invisible to the mean satisfaction score; and the American Customer Satisfaction Index dataset (n=7,341; Hult & Morgeson, 2023), where respondents with identical satisfaction scores show up to 2.3x variation in complaint rates depending on geometric subtype. The convergence of findings across employee retention and customer satisfaction demonstrates that single-axis collapse is a general structural problem, not a domain-specific one. The core hypothesis is that the GAS captures meaningful variation in emotional experience that single-axis instruments systematically miss.

1. Introduction

1.1 The Problem with Single-Axis Measurement

A customer rates a product 7 out of 10. That number carries less information than it appears to. Seven could mean calm satisfaction. Seven could mean excited enthusiasm. Seven could mean resigned acceptance. All three are different emotional states that would sit at different locations in a multidimensional affective space. A single-axis scale conflates them because the scale has one dimension and emotional experience has at least three.

This is not a calibration problem. It is a structural limitation of the instrument. A single-axis scale cannot distinguish between states that differ on axes the scale does not measure. Two respondents with identical scores can have different experiences. Two respondents with different experiences can produce identical scores. The instrument is blind to the differences it does not measure.

1.2 Existing Multi-Dimensional Approaches

The circumplex model of affect (Russell 1980) proposed two dimensions: valence and arousal. The PANAS scale (Watson et al. 1988) measures positive and negative affect as separate dimensions. The Self-Assessment Manikin (Bradley and Lang 1994) uses pictorial scales for valence, arousal, and dominance.

These instruments advanced the field beyond single-axis measurement. None achieved widespread adoption outside of research psychology. The Likert scale, the NPS, and the PHQ-9 dominate practical survey work because they

are simple. One question. One number. Easy to administer. Easy to aggregate. Easy to report.

The Geometric Affective Survey aims to approach that simplicity while providing three-dimensional resolution. A comparable number of questions. Three coordinates instead of one number. Visualization that would make the additional resolution immediately apparent.

1.3 Theoretical Basis

The GAS is built on a three-dimensional affective state space developed as a component of a synthetic cognitive architecture. The valence axis (Riggleman 2026j) defines the positive-negative dimension of emotional experience relative to a personality setpoint called sigma. The activation axis (Riggleman 2026k) defines arousal level independent of valence. The intensity axis (Riggleman 2026l) defines magnitude of emotional displacement independent of direction. The full geometry (Riggleman 2026m) unifies these three axes into a single coordinate system where emotional state is a point in space and the distance from sigma is a computed quantity that drives multiple downstream behaviors. The GAS operationalizes that theoretical construct as a measurement instrument for human respondents.

2. The Instrument

2.1 The Three Axes

Valence measures the positive-negative quality of the emotional experience. Not how much the respondent liked something. How the experience felt on the good-bad axis. This is the axis existing instruments already measure, though they usually conflate it with the other two.

Activation measures the arousal level of the emotional experience. A calm positive experience and an excited positive experience have the same valence and different activation. Existing instruments cannot tell them apart. This axis would separate serenity from excitement, both positive, and separate sadness from anxiety, both negative.

Intensity measures the magnitude of the emotional displacement. How strongly the respondent felt whatever they felt. A mild calm satisfaction and a deep calm satisfaction have the same valence and activation but different intensity. This axis would capture the difference between a shrug and a sigh.

2.2 Distinguishing Activation from Intensity

A reviewer noted that activation and intensity frequently overlap in real emotional experience: high-intensity emotions tend to be high-activation. The distinction must be operationalized clearly or the two axes risk being redundant.

Activation is behavioral readiness. Calm versus agitated. Are you sitting still or pacing the room. It is observable from the outside.

Intensity is how strongly the state is experienced. Faint versus overwhelming. Is the feeling at a whisper or a scream. It is internal magnitude.

These two can move independently:

State	Activation	Intensity
Quiet sadness	Low	High
Nervous energy	High	Moderate
Panic	High	High
Mild irritation	Low	Low
Restless boredom	Moderate	Low

Quiet sadness is the case that does the most work. The person is not agitated. They are not pacing. Activation is low. But the feeling is enormous. A single-axis model that treats activation and intensity as the same variable would place quiet sadness low on both. The GAS places it low on one and high on the other. That difference is the argument for why both axes are needed.

The GAS does not ask respondents to distinguish activation from intensity in abstract terms. It asks concrete questions that map to each axis independently. “How energized or calm do you feel?” captures activation. “How strongly are you feeling this?” captures intensity. The respondent answers two different questions. The researcher carries the distinction.

In the theoretical model (Riggleman 2026l), intensity is computed as the normalized Euclidean distance from sigma in the valence/activation plane. It is a derived variable, not an independent self-report. For the survey instrument, intensity is self-reported directly because the respondent does not have

access to their own sigma coordinates. The theoretical derivation and the survey measurement converge on the same construct from different directions. Whether self-reported intensity correlates with the theoretically derived intensity is a testable hypothesis included in the validation agenda (Section 8.3).

2.3 Sigma Establishment

Before measuring displacement, the instrument establishes a baseline. Sigma is the respondent's default emotional state. Where they normally sit in affective space. Some people's sigma is high activation. Some people's sigma is low valence. The instrument does not assume a universal baseline. It measures each respondent's home position and computes displacement from there.

Sigma is established with three baseline questions asked before the target questions. These ask the respondent to describe their typical emotional state in terms that map to the three axes. The expectation, based on preliminary observations in the Potato system (Riggleman 2026a), is that this baseline would be stable across sessions for most respondents, meaning it would only need to be established once and verified periodically.

2.4 Sigma Addresses Individual Differences

A concern raised in early review is that subjective emotional experiences vary across individuals based on personality, culture, emotional awareness, and context. The GAS addresses this through sigma rather than treating it as uncontrolled noise.

Two respondents can report the same valence and activation but have different sigmas. One person's sigma is naturally low-activation. For them, a moderate activation reading represents significant displacement from baseline. Another person's sigma is naturally high-activation. For them, the same moderate activation reading is a return toward baseline. The raw coordinates are identical. The displacement from sigma is different. The displacement is the number that matters.

Sigma is not a normalization trick. It is a measurement of where each respondent starts. Individual differences in personality, culture, and emotional style are captured by differences in sigma position. The GAS computes displacement relative to that position, which means the same coordinate means different things for different people. This is a feature of the instrument, not a limitation. It is what existing instruments cannot do because they have no baseline to compute displacement from.

2.5 Question Format

Each target question maps to all three axes simultaneously. Instead of asking how much did you like this product on a scale of 1 to 10, the GAS presents three paired bipolar slider scales. For valence: negative to positive. For activation: calm to energized. For intensity: mild to strong. Each slider produces a value normalized to a range that maps directly to geometric coordinates.

A respondent who sets all three sliders provides a point in 3D affective space. The expected interaction time per question is comparable to a single Likert item because the three sliders would be presented simultaneously and answered in a single gestural sweep. This assumption requires empirical validation.

2.6 Output Format

Each response is a coordinate triple: (valence, activation, intensity). A set of responses is a point cloud in three-dimensional affective space. The point cloud can be visualized as a 3D scatter plot, rotated, clustered, and analyzed using standard spatial statistics.

Aggregate metrics would be computed as centroids, cluster counts, dispersion measures, and displacement vectors. These would replace the single mean score of traditional instruments with a richer set of summary statistics that preserve the dimensional structure of the data.

3. Hypothesized Advantages Over Single-Axis Instruments

3.1 Distinguishing Conflated States

A Likert scale score of 4 out of 5 conflates at least four distinct states: calm satisfaction (positive valence, low activation, moderate intensity), excited enthusiasm (positive valence, high activation, high intensity), relieved acceptance (positive valence, low activation, low intensity), and nervous optimism (positive valence, high activation, moderate intensity). The GAS would place these at four different coordinates. The hypothesis is that this separation captures real variation that matters for downstream decisions.

3.2 Cluster Analysis

When 2,000 respondents rate a product on a 1-to-10 scale and the mean is 7, the product manager has limited actionable information. If 2,000 respondents produce a point cloud in 3D affective space that clusters into two groups, one at calm-positive-moderate and one at excited-positive-high, the product manager can see that the product produces two distinct experiences. Those could map to different marketing campaigns, different target demographics, different positioning strategies. The single mean hides this. The point cloud would reveal it.

3.3 Trajectory Tracking

For longitudinal applications, the GAS could track movement through affective space over time. A respondent's coordinates at session 1, session 2, session 3 would form a trajectory. The displacement vector would show direction and magnitude of change.

Existing instruments track a number going up or down. The GAS would track a point moving through three-dimensional space. The direction of movement would carry information the magnitude alone does not. Whether this additional information improves prediction of outcomes (purchase behavior, employee retention, clinical improvement) is a testable hypothesis.

4. Backward Compatibility: Decomposing Existing Instruments

4.1 The Adoption Problem

Organizations have years of existing survey data collected on single-axis scales. A new instrument that requires discarding that historical data faces a prohibitive adoption barrier. The GAS addresses this by providing a decomposition method that can extract three-axis coordinates from existing multi-item survey data retroactively.

Most real surveys do not ask a single question. The PHQ-9 asks nine. Employee engagement surveys ask twenty or thirty. Customer satisfaction surveys ask ten to fifteen. Each question implicitly touches different axes but collapses the answer onto one scale. If the underlying questions can be mapped to valence, activation, and intensity, the three-axis decomposition can be applied to existing datasets without re-surveying respondents.

4.2 Demonstration: IBM HR Analytics Dataset

To demonstrate backward compatibility, we applied the three-axis decomposition to the IBM HR Analytics Employee Attrition and Performance dataset, a publicly available dataset containing 1,470 employee records with four satisfaction-related Likert scale measures (1–4): JobSatisfaction, EnvironmentSatisfaction, RelationshipSatisfaction, and WorkLifeBalance. The dataset also includes an attrition outcome variable indicating whether each employee left the company.

The decomposition maps existing survey items to the three geometric axes. Valence is computed from JobSatisfaction and RelationshipSatisfaction, which measure the positive-negative quality of work experience. Activation is computed from the inverse of WorkLifeBalance and JobInvolvement, where low work-life balance and high job involvement indicate high arousal. Intensity is computed from the variance across all satisfaction measures plus EnvironmentSatisfaction as a magnitude amplifier, capturing how strongly each respondent feels across all dimensions.

The single-axis score for comparison is the simple mean of all four satisfaction measures, which is what a traditional survey analysis would produce.

4.3 Result 1: The Single Axis Hides Attrition Patterns

Figure 1 shows the critical finding. The left panel displays the traditional single-axis view: a histogram of mean satisfaction scores colored by attrition outcome. The stayed and left distributions overlap almost completely. The single-axis score cannot distinguish employees who will leave from those who will stay.

The right panel shows the same 1,470 employees plotted on two of the three geometric axes: valence and activation. The attrition cases (red) visibly cluster in the low valence region. The geometric view reveals a pattern the single-axis histogram buries.

Figure 1. Left: Traditional single-axis satisfaction histogram with attrition overlaid. The distributions overlap, hiding the pattern. Right: Two-axis geometric view of the same data. Attrition cases (red) cluster visibly in the low-valence region.

4.4 Result 2: Geometric Quadrants Predict Attrition

Splitting the dataset into four quadrants based on median valence and median activation produces dramatically different attrition rates from the same satisfaction data:

Geometric Quadrant	Attrition Rate	Employees
High Valence + High Activation	11.8%	747
High Valence + Low Activation	17.5%	326
Low Valence + High Activation	21.9%	274
Low Valence + Low Activation	26.0%	123

The attrition rate in the worst geometric quadrant (26.0%) is 2.2 times the rate in the best quadrant (11.8%). This variation exists within the same satisfaction scores. The single-axis mean cannot see it. The geometric decomposition reveals it from existing data without re-surveying a single employee.

4.5 Result 3: Same Score, Different Geometry, Different Outcomes

The strongest demonstration of the instrument's value examines employees with identical single-axis scores. Among 239 employees who scored exactly 2.5 on the single-axis mean, 16% left the company. Those who left had different geometric coordinates than those who stayed: lower activation and higher intensity. The single-axis score was identical. The geometric coordinates were not. The outcomes were not.

Figure 2. All 1,470 employees plotted in three-dimensional affective space (valence, activation, intensity). Green: stayed. Red: left company. Attrition cases cluster in the low-valence, high-activation region of the space. The single-axis mean collapses this three-dimensional structure into a single number.

Figure 3. Employees with identical single-axis satisfaction scores (2.5 and 2.75) plotted in geometric space. Same number on a Likert scale. Different locations in three-dimensional space. Different attrition outcomes. The geometry sees what the number cannot.

4.6 Implications for Adoption

The backward compatibility demonstration establishes three things. First, the three axes are extractable from existing multi-item survey data without new data collection. Second, the decomposition reveals variation that the single-axis summary hides. Third, that hidden variation predicts real outcomes (attrition) that the single-axis score misses.

This means organizations do not need to discard historical data to adopt the GAS. They can decompose existing datasets, verify that the three-axis view reveals actionable patterns their current analysis misses, and then transition to native three-axis collection for new surveys. The backward compatibility path eliminates the adoption barrier.

4.7 Demonstration 2: American Customer Satisfaction Index (ACSI)

4.7.1 Dataset

To test whether the backward compatibility finding generalizes beyond employee retention, we applied the three-axis decomposition to the American Customer Satisfaction Index sample dataset (Hult & Morgeson, 2023; DOI: 10.17632/64xkbj2ry5.1). The ACSI dataset contains 8,239 consumer responses across four industries (processed food, commercial airlines, internet service providers, and commercial banks) with 15 core survey items on a 10-point scale covering expectations, perceived quality, perceived value, satisfaction, complaint behavior, and loyalty intentions. After excluding records with missing values on key variables, the analysis sample is n=7,341.

4.7.2 Geometric Mapping

The decomposition maps ACSI items to the three geometric axes using the same logic as the IBM HR decomposition: group items by the dimension they most directly measure, compute axis scores, and establish sigma from the expectations baseline.

Quality axis (valence equivalent): Average of overall quality (OVERALLQ), customization quality (CUSTOMQ), and reliability quality (WRONGQ). These measure the positive-negative evaluation of the actual experience.

Value axis (activation equivalent): Average of price-given-quality (PQ) and quality-given-price (QP). These measure the engagement dimension — whether the transaction felt worth the energy.

Sigma (expectations baseline): Average of overall expectations (OVERALLX), customization expectations (CUSTOMX), and reliability expectations (WRONGX). These measure where the respondent expected to be before the experience. This is sigma: the baseline the experience is evaluated against.

The traditional single-axis score is the respondent's overall satisfaction rating (SATIS, 1–10). The geometric subtypes are defined by whether quality and value each exceeded or fell short of sigma (expectations).

4.7.3 Result 1: Same Score, Different Complaint Rates

Among respondents who gave an overall satisfaction rating of exactly 7 out of 10 (n=1,034), complaint rates vary by geometric subtype:

Geometric Subtype	n	Complaint Rate	Repurchase Intention
Exceeded Both (quality and value above expectations)	323	15.8%	7.26
Quality Up, Value Down	296	12.5%	7.12
Quality Down, Value Up	85	27.1%	7.18
Fell Short Both	330	28.5%	6.81

The complaint rate in the "Fell Short Both" subtype (28.5%) is 2.3 times the rate in the "Quality Up, Value Down" subtype (12.5%). All of these respondents gave the same satisfaction score. The single number cannot distinguish them. The geometry can.

4.7.4 Result 2: The Pattern Holds Across Score Levels

The variation within geometric subtypes is not specific to one satisfaction score. Within SATIS=6: 2.0x variation (17.8% to 36.5%). Within SATIS=7: 2.3x variation (12.5% to 28.5%). Within SATIS=8: 2.2x variation (10.3% to 22.3%). The geometry finds hidden structure at every score level where the total score is the same.

4.7.5 Result 3: The Pattern Holds Across Industries

Within the above-average satisfaction band (total score 6.7–8.0), the geometric subtypes produce different complaint rates in every industry:

Industry	n	Exceeded Both Complaint Rate	Fell Short Both Complaint Rate
Processed Food	542	2.9%	7.6%
Airlines	379	7.1%	10.8%
ISPs	478	16.4%	25.6%
Banks	608	18.8%	22.1%

The absolute complaint rates differ by industry (food customers complain less than ISP customers). But in every industry, the “Fell Short Both” subtype complains more than the “Exceeded Both” subtype within the same satisfaction band. The geometric pattern is not industry-specific.

4.7.6 The Sigma Finding

The ACSI dataset provides a direct test of sigma because the survey includes explicit expectations questions. Sigma is not inferred from other variables as in the IBM HR demonstration. It is measured directly. Respondents in the “Fell Short Both” subtype have higher expectations (mean sigma 8.80 on a 10-point scale) than those in the “Exceeded Both” subtype (mean sigma 7.46). They expected more. They got less. They gave the same overall satisfaction score. But their complaint behavior reveals the disappointment the score conceals.

This is the sigma argument in empirical form. Two respondents at the same satisfaction score with different sigmas are having different experiences. The one with higher sigma is more displaced from their baseline. The displacement predicts the complaint behavior the total score does not.

4.7.7 Cross-Dataset Convergence

The IBM HR dataset and the ACSI dataset produce structurally identical findings in different domains:

Finding	**IBM HR (n=1,470)**	**ACSI (n=7,341)**
Same score, different outcomes	2.2x attrition variation	2.3x complaint variation
What the geometry reveals	Attrition clusters in low-valence quadrant	Complaints cluster in fell-short-of-sigma subtype
Sigma role	Inferred from item mapping	Directly measured from expectations
Domain	Employee retention	Customer satisfaction

Two different datasets. Two different domains. Two different outcome variables. Two different sample sizes. The same structural finding: the single-axis score hides variation that the geometric decomposition reveals, and that hidden variation predicts real outcomes.

5. Proposed Validation Study 1: Product Experience

5.1 Study Design

Administer both a standard 1-to-10 satisfaction scale and the GAS to the same respondents evaluating the same product. The hypothesis is that respondents with identical satisfaction scores will occupy different locations in GAS space, demonstrating that the GAS captures variation the single-axis instrument misses.

5.2 Predictions

Respondents who score 7 on the satisfaction scale but differ on activation (calm vs. excited satisfaction) would cluster at different GAS coordinates. This separation would be invisible to the single-axis instrument and visible in the 3D point cloud. The prediction is falsifiable: if all respondents with the same satisfaction score also cluster at the same GAS coordinate, the additional axes add no information.

5.3 Commercial Relevance

Metric	Likert / NPS	GAS (proposed)
Output per respondent	Single number	3D coordinate
Aggregate output	Mean score	Point cloud + clusters
Distinguishes calm vs. excited satisfaction	No	Hypothesized yes
Trajectory tracking	Number goes up or down	Vector in 3D space
Backward compatible	N/A (current standard)	Yes (Section 4 demonstration)

6. Proposed Validation Study 2: Employee Satisfaction

6.1 Study Design

An employee satisfaction survey administered quarterly using both a standard engagement scale and the GAS. The hypothesis is that the GAS detects directional changes in employee emotional state that the engagement score misses.

6.2 Predictions

An employee whose engagement score stays at 7 for three quarters but whose GAS coordinates shift from calm-positive to anxious-positive would be showing a trajectory toward burnout that the engagement score cannot see. The activation axis would catch the shift that the valence-only instrument misses. Whether this trajectory predicts actual burnout, turnover, or performance decline is a testable outcome hypothesis. The IBM HR dataset analysis in Section 4 provides preliminary support: geometric quadrant membership predicted attrition rates ranging from 11.8% to 26.0% within the same overall satisfaction score range.

7. Proposed Validation Study 3: Clinical Intake

7.1 Study Design

A clinical intake questionnaire administered alongside the PHQ-9 at initial assessment. The hypothesis is that the GAS distinguishes clinical presentations that the PHQ-9 conflates.

7.2 Predictions

A PHQ-9 score of 15 describes moderate depression. Two patients with a PHQ-9 score of 15 can present very differently. One is low valence, low activation, low intensity: withdrawn, flat, disengaged. The other is low valence, high activation, high intensity: anxious, agitated, distressed. These may require

different treatment approaches. The PHQ-9 score is identical. The GAS coordinates would differ. Whether this separation predicts differential treatment response is a clinical hypothesis that a companion paper (Riggleman 2026s) proposes to test.

8. Psychometric Properties to Validate

8.1 Test-Retest Reliability

The GAS should demonstrate test-retest reliability when the same respondent evaluates the same stimulus on two occasions. The coordinate produced should be consistent within a tolerance band. Standard psychometric validation protocols (Cronbach alpha, ICC) would apply to each axis independently and to the composite 3D coordinate.

8.2 Convergent Validity

GAS valence scores should correlate with existing single-axis satisfaction measures. GAS activation scores should correlate with arousal subscales of instruments like the PANAS. GAS intensity scores should correlate with reported importance or personal relevance of the stimulus.

8.3 Discriminant Validity

The three axes should provide independent information. If activation and intensity are perfectly correlated across respondents, the three-axis model reduces to two and the third axis adds nothing. The prediction is that the axes are correlated but not redundant, each capturing variation the others do not.

9. Discussion

9.1 What This Paper Proposes and What It Demonstrates

This paper proposes an instrument and demonstrates its backward compatibility on real data. The GAS design follows from a theoretical model of three-dimensional affective state formalized across four companion papers: the valence axis (Riggleman 2026j), the activation axis (Riggleman 2026k), the intensity axis (Riggleman 2026l), and the unified geometry (Riggleman 2026m). That model has been implemented in a synthetic cognitive architecture (Riggleman 2026a) where it drives memory decay, deception thresholds, and nightmare formation.

The IBM HR Analytics demonstration in Section 4 shows that the three-axis decomposition is not merely theoretical. Applied to 1,470 real employee records, it revealed attrition patterns invisible to the single-axis summary and produced a 2.2x difference in attrition rates across geometric quadrants. This is a demonstration of backward compatibility on existing data, not a full validation of the GAS as a native instrument. The step from demonstrated decomposition to validated native instrument requires the empirical studies described in Sections 5 through 8.

The ACSI demonstration in Section 4.5 extends the backward compatibility finding to a second domain with a larger sample (n=7,341) and a directly measured sigma (expectations). The convergence of findings across employee retention and customer satisfaction — both showing approximately 2x variation in outcome rates within the same single-axis score — is evidence that the

structural limitation is general. Any survey that reduces multidimensional experience to a single scalar will hide the same kind of variation. The geometry is not domain-specific. The problem it solves is not domain-specific.

9.2 The Bridge Between Theory and Measurement

The theoretical claim in the companion papers is that emotional state is a point in three-dimensional space and the distance from sigma is a single computed quantity with predictive power. The measurement claim in this paper is narrower: that three axes capture more variation in human emotional experience than one axis does. The IBM HR demonstration supports this narrower claim. The measurement claim does not depend on the theoretical claim being fully correct. The theory motivates the instrument. The instrument stands or falls on its own psychometric properties and its ability to reveal patterns that matter.

9.3 Limitations

The GAS adds response time per item. Three sliders take longer than one circle on a Likert scale. The additional time is estimated at about 2 seconds per item but this needs empirical measurement. For surveys with many items, the cumulative burden may reduce completion rates.

The sigma establishment phase adds questions at the beginning of the survey. For one-time surveys this overhead may not be justified. For longitudinal applications where sigma is established once and reused, the overhead is amortized over many sessions.

The GAS requires a digital interface. Three simultaneous sliders cannot be rendered on a paper form in a way that preserves the gestural simplicity of the design. Most commercial and clinical survey work is already digital, so this constraint is practical rather than theoretical.

The backward compatibility decomposition in Section 4 involves mapping decisions (which existing items map to which axes) that require domain expertise and could be done differently by different researchers. The specific mapping used here is one reasonable interpretation. Other mappings could produce different results. The validation studies in Sections 5–7 would use native three-axis collection, eliminating this ambiguity.

10. Conclusion

The Geometric Affective Survey proposes to measure emotional experience in three dimensions instead of one. Two independent demonstrations show this is not hypothetical. The IBM HR Analytics dataset (n=1,470) reveals a 2.2x difference in attrition rates across geometric quadrants among employees with overlapping single-axis scores. The ACSI dataset (n=7,341) reveals up to 2.3x variation in complaint rates across geometric subtypes among customers with identical satisfaction scores. The pattern holds across score levels, across industries, and across domains.

The instrument does one proposed thing: give emotional experience a location in space instead of a position on a line. That location carries more information. The three axes are not new to affective science. The circumplex model proposed two of them in 1980. What this paper proposes is packaging them in an instrument designed to be simple enough to replace the Likert scale in practical survey work, with backward compatibility that eliminates the adoption

barrier, and with cross-domain evidence that the hidden variation is real and consequential. Whether the instrument generalizes to native three-axis collection is an empirical question the proposed validation studies are designed to answer.

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